UNIT – IV

Balancing of Rotating Masses & Cams

Subject : Theory of Machines (23A03403)

Syllabus: Need for balancing, balancing of a single mass and several masses in different planes (analytical & graphical) ; classification of cams & followers, terminology, displacement diagrams – uniform velocity, parabolic, SHM, cycloidal ; derivatives of follower motion ; specified-contour cams – circular and tangent cams ; pressure angle & undercutting.

1. Balancing – Need

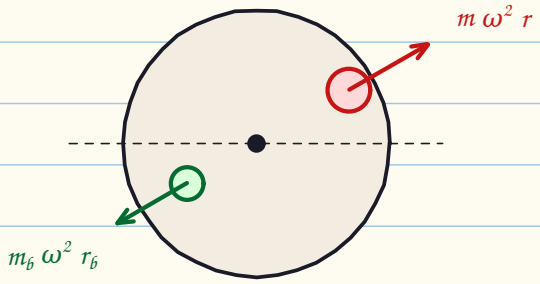
A rotating mass m at radius r with angular velocity ω produces a centrifugal force $F = m\,\omega^2\,r$ which acts radially and rotates with the shaft – it causes vibration, noise, bearing wear, fatigue failure and discomfort.

- Example : 5 kg unbalanced mass at $r = 0.1$ m, $N = 3000$ rpm : $\omega = 314.2$ rad/s $\Rightarrow F = 5 \times 0.1 \times 314.2^2 = 49.3$ kN ($\approx$ 1000 times its weight) – hence balancing of rotors of turbines, motors, crankshafts, wheels, fans.
- **Static balance** : resultant centrifugal force = 0 (the CG lies on the axis) $\Rightarrow \Sigma\, m\,r = 0$ (vector sum). **Dynamic balance** : resultant force = 0 and resultant couple = 0 $\Rightarrow \Sigma\, m\,r = 0$ and $\Sigma\, m\,r\,l = 0$. A dynamically balanced rotor is always statically balanced, not vice versa (e.g. two equal masses at opposite sides in different planes).
- Causes of unbalance : eccentric mounting, non-uniform material, keyways, bent shafts, unequal blades, wear. Balancing = adding/removing mass so that the inertia forces are equal and opposite.

2. Single Rotating Mass

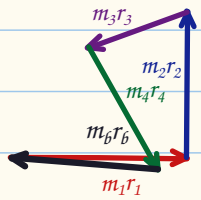
(a) Balanced by one mass in the same plane

Single mass in one plane



$m_b r_b = m r$ (opposite)

Force polygon (same plane)



closing vector = balancing mr

Mass m at radius r. Add a balancing mass m_b at radius r_b, diametrically opposite, in the same plane :

$$m_b r_b = m r$$

(b) Balanced by two masses in two other planes

A single mass m in plane A at radius r cannot be balanced by one mass in a different plane (forms a couple). Use two masses m_1, m_2 in planes 1 and 2 at distances l_1, l_2 from A :

- Force condition : $m_1r_1 + m_2r_2 = m r$ (both on the opposite side).
- Couple condition (moments about plane 2) : $m r \times l_2 = m_1r_1 (l_1 + l_2) \Rightarrow m_1r_1 = m r\ l_2/(l_1 + l_2)$; likewise $m_2r_2 = m r\ l_1/(l_1 + l_2)$ — like the reactions of a simply-supported beam. If A lies between the planes both balancing masses are opposite to m ; if A is outside, the nearer mass is on the same side as m.
- **Principle** : a mass in one plane can be replaced by an equal mass in the reference plane plus a couple (m r × distance).

3. Several Masses in the Same Plane

- Masses m_1, m_2, ... at radii r_1, r_2, ... and angles θ_1, θ_2, ... (all in one plane) ⇒ unbalanced force vector Σ m r. **Graphical** : draw the polygon of vectors m r (head-to-tail, parallel to the mass directions) ; the closing vector, reversed, is the balancing $m_b r_b$ (see figure).
- **Analytical** : resolve into components :

$\Sigma H = \Sigma\, m\, r \cos\theta$ $\Sigma V = \Sigma\, m\, r \sin\theta$ $R = \sqrt{(\Sigma H^2 + \Sigma V^2)}$, $\tan\theta_b = \Sigma V / \Sigma H$ (balance at $\theta_R + 180°$)

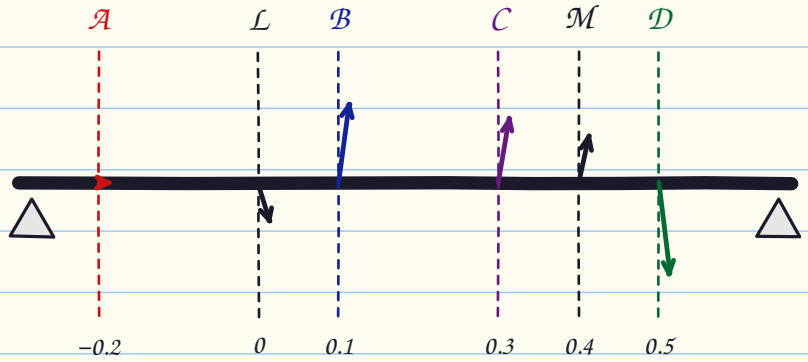
- Balancing mass $m_b = R / r_b$, placed at $\theta_b = \theta_R + 180°$.

Ex 1. Masses 15 kg at 0.2 m (0°), 10 kg at 0.25 m (90°), 12 kg at 0.15 m (200°), 8 kg at 0.3 m (300°). mr : 3.0, 2.5, 1.8, 2.4 kg·m. $\Sigma H = 3.0 + 0 - 1.69 + 1.2 = 2.51$; $\Sigma V = 0 + 2.5 - 0.62 - 2.08 = -0.19 \Rightarrow R = 2.52$ kg·m at −4.4°.

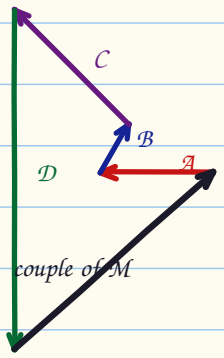
Balancing $mr = 2.52$ kg·m at 175.6° ; if $r_b = 0.2$ m $\Rightarrow m_b = 12.6$ kg.

4. Several Masses in Different Planes

Several planes : reference plane L
Planes (distances from L in m) — L, M = balancing planes



Couple polygon ($m\,r\,\chi$)



closing vector $\Rightarrow m_M r_M \chi_M$

Masses in different transverse planes along the shaft : both conditions (force & couple) must be satisfied. A reference plane (RP) is chosen ; the balancing masses are placed in two planes L and M (one of which may be chosen as RP to simplify).

Analytical procedure

- (1) Choose the reference plane (usually L) ; tabulate for each mass : m, r, θ, mr, distance l from RP (+ve to the right, −ve to left), couple mrl.
- (2) **Couple balance** : resolve mrl components : $\Sigma H_c = \Sigma\, mrl \cos\theta$, $\Sigma V_c = \Sigma\, mrl \sin\theta \Rightarrow$ the balancing couple of the mass in plane M : $m_M r_M l_M = \sqrt{(\Sigma H_c^2 + \Sigma V_c^2)}$ acting opposite to the resultant couple.
- (3) **Force balance** : include $m_M r_M$; $\Sigma H + m_M r_M \cos\theta_M + m_L r_L \cos\theta_L = 0$ and similarly for $V \Rightarrow m_L r_L$ and θ_L.
- Taking moments about the plane L eliminates the unknown mass in L.

Graphical procedure

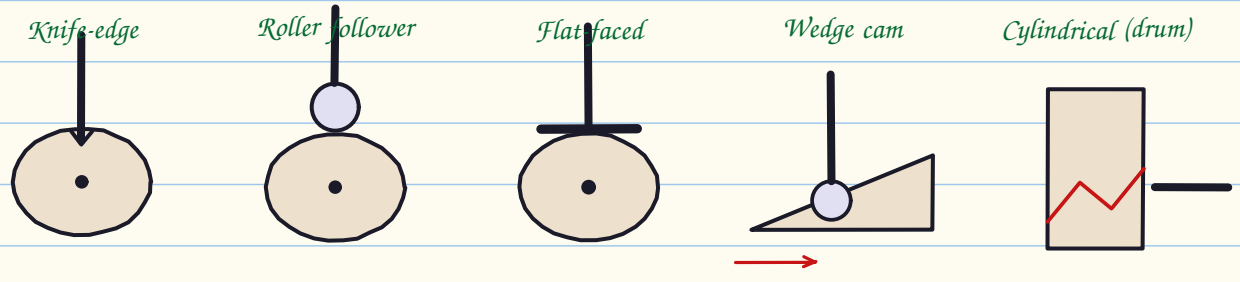
- Draw the couple polygon (vectors m r l, drawn parallel to the radii) ⇒ the closing vector gives $m_M r_M l_M$; draw the force polygon including $m_M r_M$ ⇒ closing vector gives $m_L r_L$ and its angle. (Couple vectors are drawn in the direction of the mass if l is positive, reversed if negative.)

Ex 2. A (200 kg, 0.15 m, 0°, l = −0.2 m), B (300 kg, 0.10 m, 60°, +0.1), C (240 kg, 0.12 m, 135°, +0.3), D (260 kg, 0.14 m, 270°, +0.5) about plane L ; balancing mass in plane M (l = 0.4 m, r = 0.5 m). Couples m r l : A −6.00, B 3.00, C 8.64, D 18.2 kg·m². $\Sigma H_c = -10.61$; $\Sigma V_c = -9.49$ ⇒ $|C| = 14.24$ ⇒ $m_M r_M = 14.24/0.4 = 35.6$ kg·m at 41.8° ⇒ m_M = 71.2 kg (r = 0.5). Force balance : $\Sigma H = 24.6$, $\Sigma V = 9.9$ kg·m ; adding $m_M r_M$ (26.5, 23.7) ⇒ $m_L r_L$ = 61.2 kg·m at 213.4° ⇒ m_L = 122.5 kg.

- **Remarks** : the choice of planes is limited by space ; the balancing masses are usually in the end planes of a rotor (turbine disc, armature) ; balancing machines detect the dynamic reaction at bearings. Residual unbalance is expressed as G (mm/s) grade, ISO 1940.
- **Balancing of a rotor in the field** : trial-weight method (two-plane), influence coefficients ; balancing of flexible rotors operating near critical speeds needs multi-plane balancing.
- (Reciprocating and multi-cylinder engine balancing – primary & secondary forces, in-line, V, radial – is outside this unit but follows the same force & couple method.)

5. Cams – Classification & Terminology

A cam is a rotating (or translating) machine element that gives a prescribed, usually non-uniform, motion to a follower by direct contact – a higher pair.



Classification of cams

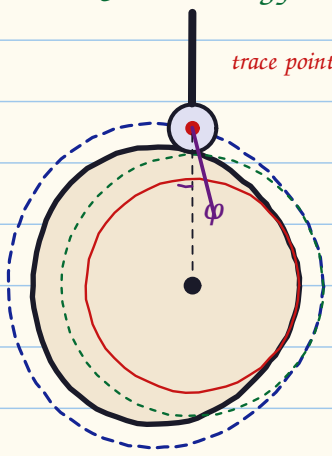
- **Radial (disc/plate) cams** – follower moves radially. **Translating (wedge)** – cam moves linearly. **Cylindrical (drum) cam** – follower moves parallel to the axis ; groove or end cam. Also conjugate, globoidal, spherical cams and cams with specified contour.

Classification of followers

Basis	Types	Remarks
Shape	Knife-edge / roller / flat-faced (mushroom) / spherical	knife-edge wears, roller common, flat for high speed
Motion	Reciprocating (translating) / oscillating (swinging arm)	valves / relay mechanisms
Line of motion	In-line (passes through cam centre) / offset	offset reduces pressure angle in rise
Closure	Spring, gravity, positive (groove, conjugate)	high speed – positive drive

Terminology

Cam terminology



Base circle (r_b) – smallest circle

Prime circle ($r_b + r_{roller}$)

Pitch curve – path of trace point

Cam profile – working surface

Pressure angle φ : normal ∧ follower path

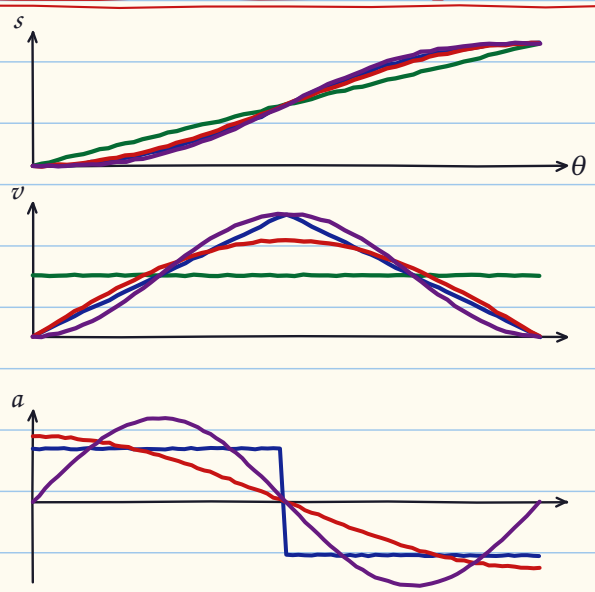
Lift (stroke) h = max s

Angles : rise δ_1 → dwell → return δ_2 → dwell

- **Base circle** : smallest circle tangent to the cam profile (radius r_b). **Trace point** : reference point on the follower (knife edge, roller centre). **Pitch curve** : path of the trace point relative to the cam. **Prime circle** : smallest circle from the cam centre tangent to the pitch curve (radius $R_p = r_b + r_{roller}$).
- **Pressure angle φ** : angle between the direction of follower motion and the common normal at the contact point. **Pitch point** : the point on the pitch curve with maximum pressure angle. **Lift (stroke) h** : maximum follower displacement.

- **Angle of ascent (rise) δ_1, dwell** (follower stationary), **angle of descent (return) δ_2,** remaining dwell. Total 360°. A displacement diagram plots s against θ (or time).

6. Follower Motions – Displacement Diagrams



Follower motions (rise h in δ)

Uniform velocity

$s = h\theta/\delta$; $v = h\omega/\delta$; $a = 0$ (∞ at ends)

Uniform acc./retardation

$s = 2h(\theta/\delta)^2$; $v_{max} = 2h\omega/\delta$; $a = \pm 4h\omega^2/\delta^2$

Simple harmonic

$s = (h/2)(1-\cos\pi\theta/\delta)$; $v_{max} = \pi h\omega/2\delta$; $a_{max} = \pi^2 h\omega^2/2\delta^2$

Cycloidal

$s = h[\theta/\delta - \sin(2\pi\theta/\delta)/2\pi]$; $v_{max} = 2h\omega/\delta$; $a_{max} = 2\pi h\omega^2/\delta^2$

Best : cycloidal (smooth a)

Cam rotates at ω ; rise h in cam angle δ (time $t = \delta/\omega$). Let θ = angle from start of rise.

(i) Uniform velocity

$$s = h\theta/\delta \quad v = h\omega/\delta \quad a = 0 \text{ (theoretically } \infty \text{ at the ends – infinite jerk/impact)}$$

- Straight-line displacement ; used only with smoothing arcs at the ends (low speed).

(ii) Uniform acceleration & retardation (parabolic)

$$0 \le \theta \le \delta/2 : s = 2h(\theta/\delta)^2 \text{ , } v = 4h\omega\theta/\delta^2 \text{ , } a = 4h\omega^2/\delta^2 \text{ (second half : retardation of equal magnitude)}$$

- Minimum possible peak acceleration for the given rise ($a_{max} = 4h\omega^2/\delta^2$, $v_{max} = 2h\omega/\delta$) but a sudden change of acceleration (finite jerk ∞ at start, middle, end) ; used for low-medium speeds.

(iii) Simple harmonic motion

$$s = (h/2)(1 - \cos(\pi\theta/\delta)) \quad v = (\pi h\omega/2\delta)\sin(\pi\theta/\delta) \quad a = (\pi^2 h\omega^2/2\delta^2)\cos(\pi\theta/\delta)$$

- $v_{max} = \pi h\omega/2\delta$ (mid-rise) ; $a_{max} = 4.93h\omega^2/\delta^2$ (start & end). Acceleration jumps from 0 to a_{max} at the ends of a dwell – step in a ⇒ moderate shock ; smooth if no

dwell (rise-return).

(iv) Cycloidal motion

$$s = h\,[\,\theta/\beta - \sin(2\pi\theta/\beta)/2\pi\,]\quad v = (h\omega/\beta)(1 - \cos(2\pi\theta/\beta))\quad a = (2\pi h\omega^2/\beta^2)\sin(2\pi\theta/\beta)$$

- $v_{max} = 2h\omega/\beta$; $a_{max} = 2\pi h\omega^2/\beta^2 = 6.28h\omega^2/\beta^2$. Acceleration is zero at the ends and varies sinusoidally – no sudden change of acceleration (finite jerk) ⇒ least vibration & wear ; best for high-speed cams.

Motion	v_{max}	a_{max}	Remark
Uniform velocity	$h\omega/\beta$	∞	impact at ends
Parabolic	$2h\omega/\beta$	$4h\omega^2/\beta^2$	lowest a_{max}
SHM	$1.57h\omega/\beta$	$4.93h\omega^2/\beta^2$	simple, a step at ends
Cycloidal	$2h\omega/\beta$	$6.28h\omega^2/\beta^2$	smoothest

Ex 3. $h = 40$ mm, rise angle $\beta = 60°$, $N = 600$ rpm ($\omega = 62.83$ rad/s, $\beta = 1.047$ rad) : uniform $v = 2.40$ m/s ; SHM $v_{max} = \pi(0.04)(62.83)/(2\times1.047) = 3.77$ m/s, $a_{max} = \pi^2(0.04)(62.83)^2/(2\times1.047^2) = 711$ m/s² ; cycloidal $v_{max} = 4.80$ m/s, $a_{max} = 905$ m/s² ; parabolic $a = 576$ m/s².

- **Derivatives of follower motion** : $s = f(\theta)$; $v = ds/dt = \omega\, ds/d\theta$; $a = \omega^2\, d^2s/d\theta^2$; jerk $j = \omega^3\, d^3s/d\theta^3$ (rate of change of acceleration – decides vibration & noise). Fall (return) motion is obtained by symmetry : same laws with θ measured from the start of return, $s = h - f(\theta)$.
- Choice : speed ↑ ⇒ move from uniform → SHM → cycloidal. Dwell-rise-dwell (DRD) ; dwell-rise-return-dwell (DRRD) ; dwell-rise-dwell-return (DRDR – most common).

7. Profile Construction & Specified-Contour Cams

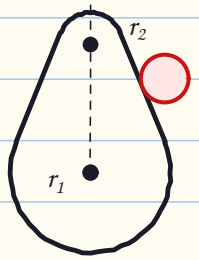
Graphical (inversion) method – steps

- (1) Draw the base circle, prime circle and the follower in its initial position. (2) Draw the displacement diagram with the same rise/dwell angles ; divide into equal parts. (3) Divide the cam circle in the same angular intervals measured opposite to cam rotation (kinematic inversion – cam fixed, follower rotating). (4) Along each radial line, mark the follower displacement from the base/prime circle.

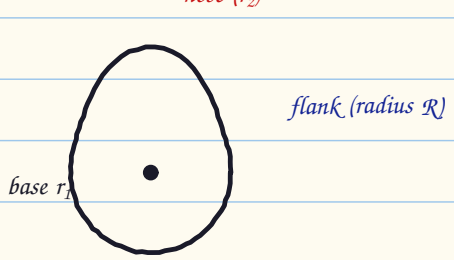
(5) Join the points – pitch curve. (6) For a roller follower, draw the roller at each point and take the envelope – cam profile (for flat-faced – draw the face line, envelope). Offset follower : lines tangent to the offset circle.

Cams with specified contours

Tangent cam (roller follower)



Circular-arc cam



flank : $s = (r_1+r_r)(\sec\theta - 1)$

nose : $x = OQ\cos\theta + \sqrt{((r_2+r_r)^2 - OQ^2\sin^2\theta)}$

- ***Tangent cam*** : the flanks are straight lines tangent to the base circle (r_1) and nose circle (r_2). For a roller follower with roller radius r_r and in-line motion, while the roller touches the straight flank the displacement from the base position :

$$s = (r_1 + r_r)(\sec\theta - 1) \quad v = (r_1 + r_r)\,\omega\,\sec\theta\tan\theta \quad a = (r_1 + r_r)\,\omega^2\,\sec\theta\,(\sec^2\theta + \tan^2\theta)$$

where θ = cam angle measured from the position when the roller starts leaving the base circle (the line of centres is perpendicular to the flank). When the roller reaches the nose circle (radius r_2) :

$$x = OQ\cos\theta + \sqrt{(r_2 + r_r)^2 - OQ^2\sin^2\theta} \quad (OQ = \text{distance between cam centre and nose centre})$$

Ex 4. Tangent cam $r_1 = 30$, $r_r = 15$ mm ($\mathcal{R}_p = 45$ mm), $N = 300$ rpm ; roller on flank at $\theta = 20°$: $s = 45(\sec 20° - 1) = 2.89$ mm ; $v = 45 \times 31.42 \times 1.064 \times 0.364 = 0.548$ m/s ; at $\theta = 30°$ $s = 6.96$ mm.

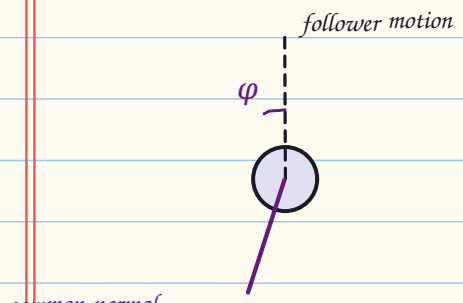
- ***Circular-arc (convex) cam*** : flanks are circular arcs of radius $\mathcal{R}$ (tangent to base circle & nose circle) – the cam has base circle r_1, nose r_2. While the roller is on the nose : $s = x - (r_1 + r_r)$ with $x = OQ\cos\theta + \sqrt{((r_2+r_r)^2 - OQ^2\sin^2\theta)}$, $OQ = r_1 + h - r_2$. On the flank the same formula with flank centre and radius $(\mathcal{R} + r_r)$.

Ex 5. Circular-arc nose : $r_1 = 30$, $h = 20$, $r_2 = 10$, $r_r = 15$ mm $\Rightarrow OQ = 40$ mm. At $\theta = 0$: $x = 40 + 25 = 65 \Rightarrow s = 20$ mm □ ; at $\theta = 20°$: $x = 40\cos 20° + \sqrt{(25^2 - 40^2\sin^2 20°)} = 37.6 + 20.9 = 58.5 \Rightarrow s = 13.5$ mm.

- Both cams are easy to manufacture (turned arcs and flats), used in automobile valve gear, but acceleration is not controlled (tangent cams give large acceleration at the nose). **Flat-faced follower** : pressure angle is zero ; contact point moves along the face by $(ds/d\theta)$ from the follower axis, so the face half-width must exceed $(ds/d\theta)_{max}$; cam size is limited by $\mathcal{R}_b + s + d^2s/d\theta^2 > 0$ (positive radius of curvature).

8. Pressure Angle & Undercutting

Pressure angle



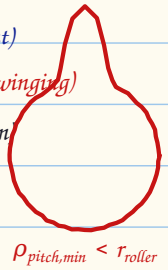
$\tan\varphi = (ds/d\theta) / (\mathcal{R}_p + s)$ (in-line)

$\tan\varphi = (ds/d\theta \mp e)/(\sqrt{(\mathcal{R}_p^2 - e^2)} + s)$ (offset)

Limits : $\varphi_{max} \approx 30°$ (translating), 45° (swinging)

↑ prime radius $\mathcal{R}_p \Rightarrow \downarrow \varphi$ (but bigger cam)

Undercutting (cusp / loop)



- **Pressure angle** φ – the angle between the common normal and the follower path. The component of the contact force along the follower = $\mathcal{F}\cos\varphi$ (useful), perpendicular component $\mathcal{F}\sin\varphi$ presses the follower sideways against its guide ⇒ friction, jamming, bending when φ is large.

$$\tan\varphi = (ds/d\theta) / (\mathcal{R}_p + s) \text{ (in-line follower, } ds/d\theta = v/\omega)$$

- Limiting values : $\varphi_{max} \approx 30°$ (reciprocating follower), 45° (oscillating). φ is largest near the middle of the rise for SHM.
- Reduction of φ : larger prime/base circle, offset the follower suitably, increase the cam rise angle, choose a favourable motion, use oscillating follower.

Ex 6. SHM rise $h = 20$ mm, $\delta = 60°$, $\mathcal{R}_p = 40$ mm : $ds/d\theta = (\pi h/2\delta)\sin(\pi\theta/\delta) = 30\sin\chi$ mm/rad ; at $\chi \approx 1.4$ rad (≈ 80° of the rise) : $s = 10(1 - \cos 1.4) = 8.3 \Rightarrow \tan\varphi = 29.5/(40 + 8.3) = 0.61 \Rightarrow \varphi = 31.4°$ (> 30° □). Increase $\mathcal{R}_p$ to 50 mm : $\tan\varphi = 29.5/58.3 = 0.506 \Rightarrow \varphi = 26.8°$ □.

Undercutting

- **Undercutting** (cusping) : the pitch curve has a radius of curvature smaller than the roller radius (convex portion), so the roller cannot trace it – the profile

develops a loop/cusp and the follower motion is altered.

$$\rho_{pitch} = [\, \mathcal{R}^2 + (d\mathcal{R}/d\theta)^2 \,]^{3/2} / [\, \mathcal{R}^2 + 2(d\mathcal{R}/d\theta)^2 - \mathcal{R}\, d^2\mathcal{R}/d\theta^2 \,] > r_r \quad (\mathcal{R} = \mathcal{R}_p + s)$$

- Avoid by : smaller roller ($r_r < 0.8\,\rho_{min}$), larger base circle, gentler motion (reduce a), limit the cam rise per angle, use a flat-faced follower only with $\mathcal{R}_b + s + d^2s/d\theta^2 > 0$.

Tabulated displacements ($h = 40$ mm, $\delta = 60°$)

θ (deg)	0	10	20	30	40	50	60
SHM s (mm)	0	2.68	10.0	20.0	30.0	37.3	40
Cycloidal s (mm)	0	1.16	7.82	20.0	32.2	38.8	40
Parabolic s (mm)	0	2.22	8.89	20.0	31.1	37.8	40

Note : cycloidal starts slowest (zero acceleration at the ends) ; parabolic has a sudden change of acceleration at the mid-point ($\theta = 30°$).

Design hints for cam–follower

- Choose the follower type by speed and load (flat or roller). Choose motion (cycloidal for high speed). Find the smallest prime circle that keeps $\varphi_{max} \le 30°$; check undercutting on the pitch curve ; offset the follower to reduce φ on the rise stroke. Specify tolerance and hardness (case-hardened, ground profile) because contact stresses are Hertzian.
- Jump phenomenon : at high speed the follower loses contact when the inertia force exceeds the spring force – spring force must exceed $m\,a_{max}$ + weight ⇒ spring stiffness $k \ge (m\,a_{max})/(\text{preload} + h)$.

10. Practical Balancing Notes

- **Balancing machines** : stand-alone (soft-bearing) and hard-bearing types measure the bearing reaction due to unbalance, then give the magnitude and angular position of correction mass in two planes.
- Correction by drilling (removing metal), adding weights, mass-adjusting screws ; for fans — trim weights on blades.

- Allowed residual unbalance $e_{per}\cdot\omega = G$ (mm/s) – for rigid rotors G = 6.3 (fans, pumps, motors), 2.5 (turbines, compressors), 1 (grinders). Example: a 12 kg rotor at 3000 rpm (ω = 314 rad/s), G = 2.5 : e_{per} = 2.5/314 = 0.008 mm ⇒ permissible unbalance U = 12 × 0.008 = 96 g·mm.

11. Additional Solved Problems

P1 (static balance). Two masses 3 kg at 0.2 m (0°) and 4 kg at 0.15 m (90°) in one plane : m r = 0.6 and 0.6 ⇒ $R = \sqrt{(0.36 + 0.36)} = 0.849$ kg·m at 45° ⇒ balancing mass 0.849 kg·m at 225° (e.g. 4.24 kg at 0.2 m).

P2 (two-plane). A mass of 10 kg at r = 0.2 m (m r = 2 kg·m) lies between planes P and Q, 0.3 m from P and 0.5 m from Q. $m_P r_P = 2 \times 0.5/0.8 = 1.25$ kg·m ; $m_Q r_Q = 2 \times 0.3/0.8 = 0.75$ kg·m ; sum 2.0 □ (both opposite to the 10 kg mass).

P3 (cam). SHM follower h = 30 mm rise in 90°, N = 120 rpm : $v_{max} = \pi h\omega/2\beta = \pi(0.03)(12.57)/(2\times1.571) = 0.377$ m/s ; $a_{max} = \pi^2 h\omega^2/2\beta^2 = 9.87(0.03)(157.9)/(2\times2.467) = 9.47$ m/s².

P4 (cycloidal). Same data : $a_{max} = 2\pi h\omega^2/\beta^2 = 6.283(0.03)(157.9)/2.467 = 12.1$ m/s².

Short answers – Quick Revision

- Why is a statically balanced rotor not necessarily dynamically balanced? The couple of the forces in different planes remains (Σ m r l ≠ 0).
- Why use cycloidal motion for high-speed cams? Acceleration starts and ends at zero – low jerk, vibration and wear.
- Effect of increasing the base circle? Smaller pressure angle, less undercutting – larger cam size.
- Which follower gives zero pressure angle? Flat-faced follower with flat perpendicular to its motion.
- Difference between pitch curve and cam profile? Pitch curve is the locus of the trace point ; the profile is offset inward by the roller radius.

12. Quick Formula Sheet – Unit IV

Item	Formula
Centrifugal force	$F = m\omega^2 r$

Item	Formula
Balance (one plane)	$\Sigma\ m\ r = 0\ ;\ m_b r_b = \mathcal{R}$
Balance (many planes)	$\Sigma\ m\ r = 0\ ;\ \Sigma\ m\ r\ l = 0$
SHM	$s = (h/2)(1-\cos\pi\theta/\beta)\ ;\ a_{max} = \pi^2 h\omega^2/2\beta^2$
Cycloidal	$a_{max} = 2\pi h\omega^2/\beta^2\ ;\ v_{max} = 2h\omega/\beta$
Parabolic	$a = 4h\omega^2/\beta^2\ ;\ v_{max} = 2h\omega/\beta$
Tangent cam	$s = (r_1+r_r)(\sec\theta - 1)$
Pressure angle	$\tan\varphi = (ds/d\theta)/(\mathcal{R}_p+s)$
Undercutting	$\rho_{pitch,min} > r_{roller}$

13. Important Questions

- Distinguish static and dynamic balancing. Explain the balancing of several masses rotating in different planes (analytical & graphical).
- Classify cams and followers with sketches. Define base circle, prime circle, pitch curve, pressure angle.
- Derive displacement, velocity and acceleration for SHM, uniform acceleration & cycloidal motion; compare them.
- Explain the construction of a cam profile for a roller follower ; describe tangent and circular-arc cams with expressions.
- What is pressure angle? How does it vary with base circle? What is undercutting and how is it avoided?